

A counter example for Lyapunov norm conjecture

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1 Introduction

Let $\mathbb{R}^{n \times n}$ denote the space of real $n \times n$ matrices, equipped with the Frobenius inner product and the induced Frobenius norm

$$\langle X, Y \rangle_F = \text{tr}(X^\top Y), \quad \|X\|_F = \sqrt{\langle X, X \rangle_F},$$

where $X \in \mathbb{R}^{n \times n}$ and $Y \in \mathbb{R}^{n \times n}$. Define the symmetric and skew-symmetric subspaces

$$\mathbb{S}^n = \{X \in \mathbb{R}^{n \times n} : X^\top = X\}, \quad \mathbb{K}^n = \{X \in \mathbb{R}^{n \times n} : X^\top = -X\}.$$

It can be proved that

$$\mathbb{R}^{n \times n} = \mathbb{S}^n \oplus \mathbb{K}^n$$

is an orthogonal direct sum.

For $A \in \mathbb{R}^{n \times n}$, define the Lyapunov operator

$$\mathcal{L}_A : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}, \quad \mathcal{L}_A(X) = AX + XA^\top.$$

Then, both $\mathbb{S}^n$ and $\mathbb{K}^n$ are invariant under $\mathcal{L}_A$:

$$\mathcal{L}_A(\mathbb{S}^n) \subseteq \mathbb{S}^n, \quad \mathcal{L}_A(\mathbb{K}^n) \subseteq \mathbb{K}^n.$$

For a linear operator $\mathcal{T}$ on $\mathbb{R}^{n \times n}$, we write

$$\|\mathcal{T}\|_F = \max_{X \neq 0} \frac{\|\mathcal{T}(X)\|_F}{\|X\|_F}$$

for the operator norm induced by the Frobenius norm.

Conjecture 1 (Lyapunov norm conjecture). *For every $A \in \mathbb{R}^{n \times n}$, the maximum*

$$\max_{X \neq 0} \frac{\|\mathcal{L}_A(X)\|_F}{\|X\|_F} = \max_{X \neq 0} \frac{\|AX + XA^\top\|_F}{\|X\|_F}$$

is attained by a symmetric matrix X . Equivalently,

$$\|\mathcal{L}_A\|_F = \|\mathcal{L}_A|_{\mathbb{S}^n}\|_F.$$

Since $\mathbb{S}^n$ and $\mathbb{K}^n$ are orthogonal invariant subspaces,

$$\|\mathcal{L}_A\|_F = \max \{ \|\mathcal{L}_A|_{\mathbb{S}^n}\|_F, \|\mathcal{L}_A|_{\mathbb{K}^n}\|_F \}.$$

Therefore Conjecture 1 is equivalent to

$$\|\mathcal{L}_A|_{\mathbb{S}^n}\|_F \geq \|\mathcal{L}_A|_{\mathbb{K}^n}\|_F, \quad (1)$$

for every $A \in \mathbb{R}^{n \times n}$.

The problem was studied by Byers and Nash [1]. Their proposed general proof contained an error, but they established several structural properties and proved the result for normal matrices. Chen and Tian [2] later corrected the status of the problem and proved the conjecture for all matrices of order at most five. Other positive cases include nonnegative, nonpositive, and tridiagonal matrices [5]. Kronecker-product formulations and computational ways of comparing the symmetric and skew-symmetric restrictions were developed in [3, 4, 5].

In this note, we give an exact 7×7 counterexample to (1), explain how it was discovered, and record simple ways to obtain further counterexamples. Some of the notations used in this note are listed below:

1. $\mathcal{L}_A$: The Lyapunov operator $X \mapsto AX + XA^\top$.
2. $\mathbb{S}^n$ and $\mathbb{K}^n$: The real symmetric and skew-symmetric $n \times n$ matrix subspaces.
3. $\|\mathcal{T}\|_F$: The operator norm of a linear map $\mathcal{T}$: induced by the Frobenius norm.
4. E_{ij} : The matrix unit whose only nonzero entry is a one in position (i, j) .
5. $\mathcal{B}_S$: The unnormalised basis of $\mathbb{S}^7$ used in the exact certificate.
6. W : The Frobenius Gram matrix of $\mathcal{B}_S$.
7. R_S : The coordinate matrix of $\mathcal{L}_A|_{\mathbb{S}^7}$ in the basis $\mathcal{B}_S$.

2 An exact 7-by-7 counterexample

Consider the integer matrix

$$A = \begin{bmatrix} 5 & 0 & -2 & 0 & 5 & 1 & 0 \\ 0 & 5 & 5 & -5 & -2 & -1 & -1 \\ 5 & 0 & 2 & 0 & 2 & 0 & -2 \\ 6 & 2 & 0 & 2 & -4 & -5 & 2 \\ -1 & -2 & 6 & 0 & 2 & 1 & -1 \\ 2 & 7 & -5 & -1 & 1 & 1 & -1 \\ -4 & 4 & 2 & 6 & 2 & -2 & -2 \end{bmatrix}. \quad (2)$$

We prove the strict separation

$$\|\mathcal{L}_A|_{\mathbb{S}^7}\|_F^2 < 309 < \|\mathcal{L}_A|_{\mathbb{K}^7}\|_F^2. \quad (3)$$

Thus A disproves the Lyapunov norm conjecture.

2.1 Exact upper bound on the symmetric restriction

To represent $\mathcal{L}_A|_{\mathbb{S}^7}$ in matrix, we use the basis

$$\mathcal{B}_S = (B_1, \dots, B_{28}) = (E_{11}, \dots, E_{77}, E_{12} + E_{21}, E_{13} + E_{31}, \dots, E_{67} + E_{76})$$

of $\mathbb{S}^7$. Let $\text{vec} : \mathbb{R}^{7 \times 7} \rightarrow \mathbb{R}^{49}$ denote column-wise vectorisation and define the basis matrix

$$P_S = [\text{vec}(B_1) \ \dots \ \text{vec}(B_{28})] \in \mathbb{R}^{49 \times 28}.$$

For $s = [s_1, \dots, s_{28}]^\top \in \mathbb{R}^{28}$, the corresponding matrix $S = \sum_{j=1}^{28} s_j B_j$ satisfies $\text{vec}(S) = P_S s$. And if we define $W = P_S^\top P_S = \text{diag}\{I_7, 2I_{21}\}$, the norm of S can be represented by s in

$$\|S\|_F^2 = \|s\|_W^2 := s^\top W s.$$

On the other hand, the Lyapunov operator (on $\mathbb{R}^{7 \times 7}$) has the Kronecker-product representation

$$\mathbf{L}_A = I_7 \otimes A + A \otimes I_7 \in \mathbb{R}^{49 \times 49}, \quad \text{vec}(\mathcal{L}_A(X)) = \mathbf{L}_A \text{vec}(X).$$

Since $\mathbb{S}^7$ is invariant under $\mathcal{L}_A$, there is a unique matrix $R_S \in \mathbb{Z}^{28 \times 28}$ such that

$$\mathbf{L}_A P_S = P_S R_S.$$

Consequently,

$$\|\mathcal{L}_A(S)\|_F^2 = \|\mathbf{L}_A P_S s\|_F^2 = \|P_S R_S s\|_F^2 = s^\top R_S^\top W R_S s,$$

and, for every nonzero $S \in \mathbb{S}^7$,

$$\frac{\|\mathcal{L}_A(S)\|_F^2}{\|S\|_F^2} = \frac{s^\top R_S^\top W R_S s}{s^\top W s}.$$

Define

$$M = 309W - R_S^\top W R_S.$$

It can be proved (e.g., by checking the leading principal minors of M) that M is positive definite. Hence, for every nonzero $S \in \mathbb{S}^7$, we have

$$\|\mathcal{L}_A(S)\|_F^2 < 309 \cdot \|S\|_F^2,$$

and therefore

$$\|\mathcal{L}_A|_{\mathbb{S}^7}\|_F^2 < 309. \tag{4}$$

2.2 Exact skew-symmetric witness

Take

$$K = \begin{bmatrix} 0 & -16 & -1 & 2 & 1 & 2 & 1 \\ 16 & 0 & -3 & -2 & 1 & 1 & -1 \\ 1 & 3 & 0 & 9 & -1 & 0 & -1 \\ -2 & 2 & -9 & 0 & 1 & 0 & 2 \\ -1 & -1 & 1 & -1 & 0 & -5 & 1 \\ -2 & -1 & 0 & 0 & 5 & 0 & 0 \\ -1 & 1 & 1 & -2 & -1 & 0 & 0 \end{bmatrix} \in \mathbb{K}^7.$$

A direct integer calculation gives

$$\|K\|_F^2 = 794, \quad \|AK + KA^\top\|_F^2 = 246134,$$

therefore

$$\|\mathcal{L}_A|_{\mathbb{K}^7}\|_{\mathbb{F}}^2 > \frac{\|\mathcal{L}_A(K)\|_{\mathbb{F}}^2}{\|K\|_{\mathbb{F}}^2} = \frac{246134}{794} > 309. \quad (5)$$

Combining (4) and (5) proves (3).

To check the correctness numerically, we provide a MATLAB code at:

<https://github.com/IzayoiYuuki/ACounterExampleForLyapunovNormConjecture>. All quantities in this counter example are integers. Thus the verification can be carried out exactly, without numerical error.

3 How to construct counter examples for $n > 7$

Let

$$\hat{A} = \text{diag}\{A, I_{n-7}\}, \quad n > 7.$$

For $A_1 \in \mathbb{R}^{p \times p}$ and $A_2 \in \mathbb{R}^{q \times q}$, define

$$\mathcal{L}_{A_1, A_2} : \mathbb{R}^{p \times q} \mapsto \mathbb{R}^{p \times q}, \quad \mathcal{L}_{A_1, A_2}(Z) = A_1 Z + Z A_2^\top.$$

A block decomposition of symmetric and skew-symmetric matrices gives

$$\begin{aligned} \|\mathcal{L}_{\hat{A}}|_{\mathbb{S}^n}\|_{\mathbb{F}} &= \max \{ \|\mathcal{L}_A|_{\mathbb{S}^p}\|_{\mathbb{F}}, \|\mathcal{L}_{I_{n-7}}|_{\mathbb{S}^q}\|_{\mathbb{F}}, \|\mathcal{L}_{A, I_{n-7}}\|_{\mathbb{F}} \}, \\ \|\mathcal{L}_{\hat{A}}|_{\mathbb{K}^n}\|_{\mathbb{F}} &= \max \{ \|\mathcal{L}_A|_{\mathbb{K}^p}\|_{\mathbb{F}}, \|\mathcal{L}_{I_{n-7}}|_{\mathbb{K}^q}\|_{\mathbb{F}}, \|\mathcal{L}_{A, I_{n-7}}\|_{\mathbb{F}} \}. \end{aligned}$$

A direct integer calculation gives

$$\|\mathcal{L}_{A, I_{n-7}}\|_{\mathbb{F}}^2 = \|A + I_7\|_2^2 \leq \|(A + I_7)^\top (A + I_7)\|_\infty = 184 < 309.$$

Together with (4)–(5), this shows

$$\|\mathcal{L}_{\hat{A}}|_{\mathbb{S}^n}\|_{\mathbb{F}}^2 < 309 < \|\mathcal{L}_{\hat{A}}|_{\mathbb{K}^n}\|_{\mathbb{F}}^2.$$

Hence $\hat{A}$ is an integer counterexample for every $n > 7$.

4 Exploration of the case $n = 6$

The preceding sections give counterexamples for every $n \geq 7$. We now turn to the case $n = 6$ and derive necessary conditions for a counterexample.

Throughout this section, $A \in \mathbb{R}^{6 \times 6}$, and we retain the notation of the introduction. Write

$$s = s(A) = \|\mathcal{L}_A|_{\mathbb{S}^6}\|_{\mathbb{F}}, \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{15} \geq 0$$

for the symmetric operator norm and the singular values of $\mathcal{L}_A|_{\mathbb{K}^6}$, respectively. The order-6 Lyapunov norm conjecture is equivalent to $\sigma_1 \leq s$.

Lemma 1. *For every $Z \in \mathbb{K}^6$ with $\text{rank } Z \leq 4$, there is $P \in \mathbb{S}^6$ such that*

$$P^2 = -Z^2, \quad \|\mathcal{L}_A(P)\|_{\mathbb{F}} \geq \|\mathcal{L}_A(Z)\|_{\mathbb{F}}.$$

Consequently, $\|\mathcal{L}_A(Z)\|_{\mathbb{F}} \leq s\|Z\|_{\mathbb{F}}$.

Proof. We follow Chen–Tian’s proof of the case $n = 4$ in [2, Theorem 0.3]. An orthogonal change of coordinates puts Z in the form

$$Z = \text{diag}\{k_1 J_0, k_2 J_0, 0_2\}, \quad J_0 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Moreover, we can assume $a_{14} = a_{23} = 0$, because svd of $A(1 : 2, 3 : 4)$ can be done while preserving Z ’s structure. Set

$$P = \text{diag}\{k_1 F_0, k_2 F_0, 0_2\}, \quad F_0 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Then $P^2 = -Z^2$ and $\|P\|_F = \|Z\|_F$. For $U = P + Z = 2k_1 E_{12} + 2k_2 E_{34}$, the identity in the proof of [2, Lemma 0.2] gives

$$\text{tr}(\mathcal{L}_A(U)^2) = \|\mathcal{L}_A(P)\|_F^2 - \|\mathcal{L}_A(Z)\|_F^2.$$

Factor $U = XY^\top$, where $X = [2k_1 e_1, 2k_2 e_3]$ and $Y = [e_2, e_4]$. Since $Y^\top X = 0$, expansion and cyclicity of the trace yield

$$\begin{aligned} \text{tr}(\mathcal{L}_A(U)^2) &= \text{tr}((Y^\top A X)^2) + \text{tr}((Y^\top A^\top X)^2) \\ &= 4k_1^2(a_{12}^2 + a_{21}^2) + 4k_2^2(a_{34}^2 + a_{43}^2) \geq 0. \end{aligned}$$

Thus $\|\mathcal{L}_A(Z)\|_F \leq \|\mathcal{L}_A(P)\|_F \leq s\|P\|_F = s\|Z\|_F$. $\square$

Lemma 1 tell us that any counterexample in the case $n = 6$ must arise from a nonsingular skew-symmetric matrix. Furthermore, with the help of Lemma 1, we can obtain the following theorem.

Theorem 2 (Bound on the second skew singular value). *For every $A \in \mathbb{R}^{6 \times 6}$,*

$$\boxed{\sigma_2 \leq s.}$$

If $\sigma_1 > s$, then σ_1 is simple, and both matrices in its skew singular pair are invertible.

Proof. Every real two-dimensional subspace of $\mathbb{K}^6$ contains a nonzero singular matrix. To see this, choose a Frobenius-orthonormal basis Z_1, Z_2 and consider

$$Z(\theta) = \cos \theta Z_1 + \sin \theta Z_2.$$

For a skew matrix of order six, the Pfaffian is a homogeneous cubic polynomial satisfying $\det(Z) = \text{pf}(Z)^2$. Hence

$$\text{pf}(Z(\theta + \pi)) = -\text{pf}(Z(\theta)).$$

Continuity gives a zero of the Pfaffian. The corresponding matrix has norm one and is singular. Its rank is at most four, since skew matrices have even rank.

Suppose $\sigma_2 > s$. Put $\mathcal{T} = \mathcal{L}_A|_{\mathbb{K}^6}$ and choose orthonormal right singular matrices K_1, K_2 for σ_1, σ_2 . For $Z = aK_1 + bK_2 \neq 0$,

$$\|\mathcal{T}Z\|_F^2 = \sigma_1^2 a^2 + \sigma_2^2 b^2 \geq \sigma_2^2 \|Z\|_F^2 > s^2 \|Z\|_F^2.$$

Their span contains a nonzero matrix of rank at most four, contradicting Lemma 1. Thus $\sigma_2 \leq s$.

If $\sigma_1 > s$, this bound makes σ_1 simple. Its unit skew singular matrices K, N satisfy

$$\mathcal{L}_A(K) = \sigma_1 N, \quad \mathcal{L}_{A^\top}(N) = \sigma_1 K, \quad \|K\|_F = \|N\|_F = 1. \quad (6)$$

If K were singular, Lemma 1 would give $\sigma_1 \leq s$. Applying the same lemma to $A^\top$ excludes singular N , because $s(A^\top) = s(A)$. $\square$

To state the next result, write

$$\mathcal{C}(K, N) = \{R \in \mathbb{R}^{6 \times 6} : RK = KR, RN = NR\}.$$

This joint commutant always contains $\mathbb{R}I$, the scalar matrices. The condition $\mathcal{C}(K, N) \neq \mathbb{R}I$ therefore means that K, N have a nonscalar common commuting matrix.

Theorem 3 (Exclusion by common symmetries). *Let K, N be a unit skew singular pair as in (6). Then $\sigma_1 \leq s$ if either*

(a) $\mathcal{C}(K, N) \neq \mathbb{R}I$; or

(b) *there is a nonzero symmetric matrix H such that $HK = -KH$ and $HN = -NH$.*

Proof. Lemma 1 handles singular K or N , so assume both are invertible. If a nonscalar symmetric matrix commutes with K, N , one of its proper spectral projections gives a common orthogonal splitting $K = K_1 \oplus K_2$, $N = N_1 \oplus N_2$ of sizes two and four. These sizes follow because the blocks are invertible and skew-symmetric. Embed each K_i by zeros as $\hat{K}_i$. By (6) and Lemma 1,

$$\sigma_1^2 \|N_i\|_F^2 \leq \|\mathcal{L}_A(\hat{K}_i)\|_F^2 \leq s^2 \|K_i\|_F^2,$$

and summing gives $\sigma_1 \leq s$.

We may therefore assume that every symmetric matrix in $\mathcal{C}(K, N)$ is scalar. Under condition (a), closure of $\mathcal{C}(K, N)$ under transpose gives a nonzero skew member V . Since $-V^2$ is symmetric and belongs to the commutant, $-V^2 = c^2 I$ for some $c > 0$. Thus $T = V/c$ is orthogonal, skew-symmetric, and commutes with K, N . Under condition (b), H^2 belongs to the commutant, so $H^2 = c^2 I$ for some $c > 0$. Then $T = H/c$ is orthogonal, symmetric, and anticommutes with K, N .

In either case, $P = TK$ and $Q = TN$ are symmetric unit matrices with $PQ = -KN$. Hence

$$\langle Q, \mathcal{L}_A(P) \rangle_F = 2 \operatorname{tr}(APQ) = -2 \operatorname{tr}(AKN) = \langle N, \mathcal{L}_A(K) \rangle_F = \sigma_1.$$

Therefore $s \geq \|\mathcal{L}_A(P)\|_F \geq \sigma_1$. □

Together, the two theorems show that any strict counterexample must satisfy

$$\sigma_1 > s \geq \sigma_2, \quad \det K \neq 0, \quad \det N \neq 0, \quad \mathcal{C}(K, N) = \mathbb{R}I,$$

and admit no nonzero symmetric matrix anticommuting with both K and N . These are necessary conditions for a counterexample.

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